

OSF: Inferring Presentational Goals and the Welfare Trade-off Ratio Under Uncertainty

Description

We use a computational approach to examine how people make inferences under uncertainty about the welfare trade-off ratio and presentational goals when observing sharing behavior.

Hypotheses

We are broadly interested in exploring how people make inferences under uncertainty about the motivations of others when observing their behavior. More specifically, we are interested in the inferences people make about a person's welfare trade-off ratio (how much someone weighs someone's welfare relative to their own). Prosocial behavior can be motivated by care for the recipient of the action (Quillien, 2023). People can make inferences through inverse planning, whereby they work backwards from observable behavior to understand the intentions of the actor (Baker et al., 2009).

This study will test a computational model of how people make inferences under uncertainty in a social context. We present the formal details of the model in the next sub-section; here we provide the main intuition about how the model works.

Our model assumes that people use a generative model to predict and interpret the actions of other people. According to this generative model, an actor's decision is driven by the desire for monetary gains but also concern for the recipient of their actions.

We assume that people 'invert' this generative model using Bayesian inference to infer information about the actor, like the actor's concern for the recipient, on the basis of the actor's actions. Assuming that people make inferences that are approximately rational, we can explore how people use observable cues (the situation as well as the action) to infer the most likely motivations that drove the behavior of the actor.

We will test the predictions of our computational model by conducting a behavioral experiment in which participants are shown a series of situations in which an actor shares a hypothetical amount of money with them. We manipulate how much money participants will expect the actor to receive as well as knowledge about the amount of money the actor receives. We refer to the money the actor receives as their endowment. We ask participants to provide judgments about the motivations of the actor when observing their actions. In doing so, we can assess the extent to which people use inverse planning to infer the welfare trade-off ratio in uncertain situations.

We have the following hypotheses, motivated by our computational model:

- (1) Participant judgments about the actor's welfare trade-off ratio will be correlated with the model predictions about the actor's welfare trade-off ratio.
- (2) Controlling for the amount of money shared, a prior belief about the giver having a high endowment will have a negative main effect on judgments about the welfare

trade-off ratio. Participants will give lower judgments of the welfare trade-off ratio when they believe the giver is likely to be rich relative to when they believe the giver is likely to be poor.

- (3) Controlling for the amount of money shared, knowing the giver has a high endowment will have a negative main effect on judgments about the welfare trade-off ratio. Participants will give lower judgments of the welfare trade-off ratio when they know the giver is rich relative to when they know the giver is poor.
- (4) Controlling for prior beliefs about endowment and the actual endowment, sharing money will have a positive main effect on judgments about the welfare trade-off ratio. Specifically, judgments about the welfare trade-off ratio will increase with respect to the amount of money shared.
- (5) There will be an interaction between knowledge about the endowment and the amount of money the giver shares. More specifically, when the participant does not have knowledge about the givers' actual endowment, their judgments about the welfare trade-off ratio will be lower when the giver shares \$25 versus when the giver shares \$15. Conversely, when the participant has knowledge about the giver's actual endowment, their judgments about the welfare trade-off ratio will be higher when the giver shares \$25 versus when the giver shares \$15. Note that we will test this interaction in the *poor-prior* condition (in the *rich-prior* condition, our model makes less confident predictions about whether there will be an interaction).
- (6) There will be a positive correlation between judgments about empathy and judgments about the welfare trade-off ratio. As judgments of empathy increase so will judgments about the welfare trade-off ratio.

Computational Model

Formally, we assume the giver makes decisions according to the following utility function:

$$U(a; \lambda, E) = (E - R(a)) + \lambda R(a) \quad (1)$$

Where E is the endowment of the actor, with $E(\text{poor}) = 0.2$ and $E(\text{rich}) = 1.00$, $R(a)$ is the benefit to the recipient, $(E - R(a))$ is the benefit to the actor, and λ is the welfare trade-off ratio. The possible actions, the percentage of the actual endowment the giver can share, are $(0, 0.15, 0.25, 0.50, .75, 1.00)$.

The observer assumes that Alice chooses an action using a softmax with inverse temperature β over the utility function:

$$p(a|\beta, \lambda, E) \propto \exp\{\beta \cdot U(a; \lambda, E)\} \quad (2)$$

As the observer is not always sure about Alice's endowment, they hold a prior over E . This prior is determined by passing the objective probability $p(E)$ through a probability weighting function with parameter α . Thus

$$w(E; \alpha) = \exp\{-\log(2)^{1-\alpha}\{-\log p(E)\}^\alpha\} \quad (3)$$

where $\log(2)^{1-\alpha}$ ensures that $w(E = 1/2) = 1/2$. When $\alpha > 1$, probabilities below $1/2$ are under weighted, otherwise probabilities over $1/2$ are over weighted. The reverse is true when $\alpha \in [0, 1]$ (Prelec, 1998). Finally, the observer has a prior $p(\lambda|\mu, \sigma)$ over Alice's welfare tradeoff ratio. Given some action a , the observer can thus update their beliefs a obtain a joint distribution over λ and E using Bayes' rule:

$$p(\lambda, E|a; \beta, \alpha, \mu, \sigma) \propto \exp\{\beta \cdot U(a; \lambda, E)\} w(E; \alpha) p(\lambda; \mu, \sigma) \quad (4)$$

By marginalising out E , the observer can infer λ . This model has 4 free parameters. First, an inverse temperature parameter β represents the rationality of the actor. Second, there is a parameter α for a probability weighting function applied to the perceived probability of the endowment that the actor receives. Third and fourth, we have parameters of $p(\lambda)$, the prior distribution over Alice's welfare tradeoff ratio, which we may represent as a normal distribution or a skewed Laplace distribution (with location 0) following previous research (Quillien, Tooby & Cosmides, 2023).

We will fit the model jointly for all participants to model the distribution of responses rather than individual trajectories. The fitting procedure will be carried out using the NumPro MCMC library.

References:

Baker, C. L., Jara-Ettinger, J., Saxe, R., & Tenenbaum, J. B. (2009). Action understanding as inverse planning. *Cognition*, 113(3), 329-349. doi: 10.1016/j.cognition.2009.07.005

Btesh, V., Lagnado, D., & Gerstenberg, T. (2025). Taking others for granted: Balancing personal and presentational goals in action selection. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 47(0).

Quillien, T. (2023). Rational information search in welfare tradeoff cognition. *Cognition*, 231, 105317.

Prelec, D. (1998). The Probability Weighting Function. *Econometrica*, 66(3), 497–527.

Study Design

This study is a mixed design:

Between-Subjects: We manipulate participant's expectation about how much money the actor is likely to receive by showing them a gumball machine with 10 balls valued at (\$100) and (\$20). Participants will be assigned to a gumball machine with either a higher proportion of rich balls (\$100) to poor balls (\$20) or a lower proportion of rich ball balls to poor ball. We define this proportion as 8:2, where there will be two balls of one value (either rich or poor) and 8 balls of the other value (either rich or poor)

Within-Subjects: 2 x 2 x 5 design

- Actor receives either a high value ball (\$100) or a low value ball (\$20).
- The participant Sees Ball/ Does not see Ball: the participant will either be shown a ball with a question mark and told they do not know which ball was selected or they will be shown which ball the actor received.
- Donation percentage (0, .15, .25, .50, .75, 1.00), this is the proportion of the endowment that the actor shares. (Note that participants are not told the percentages directly and just see the raw monetary amount).

Participants are first shown experimental instructions and walked through an example trial. Participants are shown a total of 20 trials, where they see each possible donation percentage per each combination of conditions. In the experiment, participants are first shown either a rich or poor gumball machine (between-subjects), they are then either shown which ball the actor receives or not shown which ball the actor received, the actor

can either receive a poor or rich ball, finally participants are shown how much money the actor decided to share with them and asked to answer two questions:

- 1. How much do you think the actor cares about you? (this corresponds to asking about the welfare trade-off ratio)
- 2. How much empathy do you think the actor has for you?

The value of the balls and their associated color is counterbalanced across participants.

Randomization

Participants are randomly assigned to either the mostly-rich gumball machine condition or the mostly-poor gumball machine condition. Participants are shown the amount of money that has been shared with them by the actor. The amount of money shared is calculated as a percentage of the ball the actor receives (endowment). There are 5 possible money sharing percentages (0%, 15%, 25%, 50%, 75%, 100%). The amount shared is calculated for each type of ball received (either worth \$20 or worth \$100). Participants will see all possible percentages of money shared for both the condition where they see which ball the actor received and for when they do not see which ball the actor received. The order of questions has been randomized between participants as well as the monetary value associated with the ball color. Additionally, for each participant, the names of the actors in each combination of conditions has been randomly sampled without replacement. Finally, the order in which each combination of conditions is presented is randomized within-participant.

Data Collection Procedures

We will recruit participants online from Prolific Academic. We will screen for participants who are fluent in English, over 18, based in either the UK or US, and using a desktop (no tablets or phones).

Sample Size

Our target sample size is $N = 140$ participants.

Sample Size Rationale

We power our experiment with the view of detecting the interaction described in Hypothesis (5) described above. We will test this (within-within) interaction in the poor-prior condition. Using the Superpower package for R, we calculated that fifty participants give us 97% power to detect a medium-size interaction (Cohen's $f=.25$) in a repeated-measures design. Anticipating some possible data exclusion, we aim to recruit 70 participants in the poor-prior condition. By symmetry we will also recruit 70 participants in the rich-prior condition, for a total sample of 140 participants.

Stopping Rule

Exhaustion of monetary resources for participant recruitment or when 140 participants are recruited.

Manipulated Variables

We manipulate observing the ball selected by the giver (knows ball or does not know the ball selected), the prior probability of a given endowment (proportion of one ball color to the other ball color), the percentage of the endowment shared, and the endowment (either poor or rich).

Measured Variables

We measure participants' judgments about:

- 1. How much the agent cares about you (equivalent to the welfare trade-off ratio)
- 2. How much empathy the agent has for you

All of these questions are asked on a 7-pt likert scale (1 = none and 7 = very much) Additionally, we gather demographic information about both the age and gender of the participant.

Indices

We will compute an average response (across participants) for the welfare trade-off ratio and empathy for each combination of conditions for comparison to our model.

Statistical Models

- We will use a Pearson's correlation to compare participants' averaged responses in each condition to the predictions made by our model.
- We will use a Pearson's correlation to compare participants' empathy judgments with their welfare trade-off ratio judgments.
- We will also use a multilevel model to look at the effect of endowment prior, information about the endowment, and percentage of endowment shared on judgments about the welfare trade-off ratio.
- We will also use a multilevel model to look at the interaction between information about the endowment and the percentage of endowment shared on judgments of the welfare trade-off ratio. This model will be restricted to when the amount shared is either \$15 or \$25 dollars.

Transformations

Inference Criteria

Data Exclusion

Participants who complete the experiment in less than 4 minutes or over 30 minutes will be excluded. Participants who fail both of the comprehension questions (what are the colors of the balls in the gumball machine? and does the actor receive the ball at random?) will be excluded. Additionally, participants who provide no variance in their answers, exit the survey before complete (i.e. incomplete datasets), or fail a question designed to catch bots (question containing instructions in white-text that only a bot could read) will be excluded from the analysis.

Missing Data

Due to the design of the study, skipping a question and not providing an answer is not possible. This results in no missing data. Participants that exit before complete the experiment will be excluded from analysis.

Exploratory Analysis

- We are interested in also investigating if there is a three-way interaction between the endowment prior, the endowment of the giver, and the amount of money shared on judgments about the welfare trade-off ratio.
- We also want to run the same statistical models described above but with empathy as the dependent variable instead of the welfare trade-off ratio.
- We might also explore if an alternative computational model (see below) utilizing a joint inference about both presentational goals (how the actor wants to be perceived) and the welfare trade-off ratio is more highly correlated with participant judgments about the welfare trade-off ratio compared to our main model, which only makes an inference about the welfare trade-off ratio.

Alternative Model: Formally, we assume the giver makes a decision according to:

$$U(a) = (E - R(a)) + \lambda R(a) + \delta D_{KL}[p(\lambda | a) \parallel \rho(\lambda)] \quad (5)$$

where E , $R(a)$, E , and λ are defined as in the main model. Here, $\rho(\lambda)$ is the social target of what the actor wants others to believe about their welfare trade-off ratio (i.e. $p(\lambda|a)$), see Btesh et al. (2025) for a formal definition.

The model has 6 free parameters. First, a inverse temperature parameter β representing the rationality of the actor. Second, a parameter α for a probability weighting function applied to the perceived probability of the endowment that the actor receives. Third and fourth, parameters of $p(\lambda)$, the prior distribution over Alice’s welfare tradeoff ratio, which we may represent as a normal distribution or a skewed Laplace distribution following previous research. Lastly, fifth and sixth will be two more parameters for $\rho(\lambda)$, that is the mean and standard deviation of the target welfare tradeoff ratio the actor wants the recipient to believe.

We will fit the model jointly for all participants to model the distribution of responses rather than individual trajectories. The fitting procedure will be carried out using the NumPyro MCMC library.